

DevOps Engineering Foundation - 6-Week Intensive Bootcamp

WEEK 4: CI/CD Pipelines & Automation

Learning Outcomes:

- Understand Continuous Integration and Continuous Deployment concepts
- Build automated pipelines using GitHub Actions
- Implement automated testing in CI/CD workflows
- Deploy applications automatically using pipelines

Resources:

CI/CD Concepts:

- [DevOps CI/CD Explained in 100 Seconds](#) by Fireship (2 min)
- [CI/CD Pipeline Explained](#) by IBM Technology (8 min)
- [CI/CD Best Practices](#)

GitHub Actions:

- [GitHub Actions Tutorial - Basic Concepts and CI/CD Pipeline](#) by TechWorld with Nana (1 hour)
- [GitHub Actions Complete Course](#) by DevOps Directive (2.5 hours)
- [GitHub Actions Documentation](#)
- [GitHub Actions Marketplace](#)

Jenkins (Alternative CI/CD):

- [Jenkins Tutorial for Beginners](#) by edureka (1 hour)
- [Jenkins User Documentation](#)

Testing in CI/CD:

- [Automated Testing Tutorial](#) by freeCodeCamp (1.5 hours)

Week 4 Project:

Automated CI/CD Pipeline for Containerized Application

Build a complete CI/CD pipeline:

1. Use your Week 3 application (or create a new one)
2. Implement GitHub Actions workflows that:
 - Trigger on push and pull requests
 - Run automated tests (unit tests, linting)
 - Build Docker images on successful tests
 - Tag images with version numbers and git commit SHA
 - Push images to Docker Hub
 - Deploy to AWS EC2 instance automatically
 - Send notifications on success/failure (Slack/Email)
3. Implement staging and production environments
4. Add rollback capability
5. Include build badges in README

Deliverables:

- GitHub repository with complete workflow files
- Live application deployed via CI/CD
- Pipeline status badges
- Documentation of pipeline stages
- Video demonstration of commit → automated deployment
- Post-deployment verification tests